Algebraic Aspects of the Polynomial Littlewood–Offord Problem

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ISTA

Joint work with Matthew Kwan, Lisa Sauermann and Yiting Wang

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- Counting $\Leftrightarrow$ Probability.

Erdős–Littlewood–Offord theorem

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For independent Rademacher random variables $x_1, \ldots, x_n \in \{\pm 1\}$, how likely does $\sum_{i=1}^{n} a_i x_i + c = 0$ hold?

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If $a_1, \dots, a_n \in \mathbb{R} \setminus \{0\}, c \in \mathbb{R}$, *then*

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- Foundational result in anti-concentration theory.

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- Useful in random matrix theory etc.

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  - Meka–Nguyen–Vu: as long as $f \not\approx 0$, $\mathbb{P}[f = 0] \leq n^{\varepsilon - 1/2}$.

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Conjecture (Costello 2013, corrected)

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$$\begin{aligned} &f(\boldsymbol{x}_I, \boldsymbol{x}_J^{(1)}) - f(\boldsymbol{x}_I, \boldsymbol{x}_J) \\ &= 2 \cdot \boldsymbol{x}_I^T A_{I,J} \big(\boldsymbol{x}_J^{(1)} - \boldsymbol{x}_J\big) + \varphi\big(\boldsymbol{x}_J, \boldsymbol{x}_J^{(1)}\big). \end{aligned}$$

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Show: $\mathbb{P}\big[f(\boldsymbol{x}) = 0\big] \leq n^{\varepsilon - 1}$ or A can be made low rank by changing $o(n^2)$ entries.

- How to reduce it to the linear case?

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It is linear in $\boldsymbol{x}_I$!

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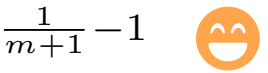
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  $\Rightarrow$ Almost all 1000×1000 of A are degenerate.
- A local-to-global phenomenon on the rank: A can be made rank-$O(1)$ up to changing $o(n^2)$ entries.

Future Directions

- So far, the proofs have been mostly combinatorial. A more geometric method to study $\{-1,1\}^n \cap \{\boldsymbol{x} : f(x_1,\ldots,x_n) = 0\}$?

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- Prove Costello's conjecture (corrected).
- What happens when $\mathbb{P}[f = 0] \gg n^{-100}$?

Thank you!

A counterexample to Costello's original conjecture

Conjecture (Costello 2013)

Suppose f is a degree-d polynomial. If f is 'far' from being reducible, then $\mathbb{P}\big[f(x_1, \ldots, x_n) = 0\big] \leq n^{\varepsilon - d/2}$.

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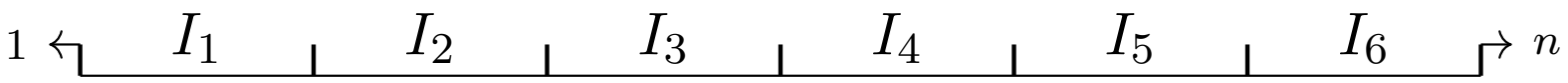
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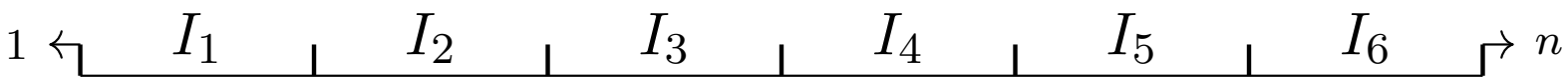
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Question

Does $\mathbb{P}[f=0] > n^{\varepsilon-d/2}$ imply that f is essentially determined by $O(1)$ polynomials of degree at most $d-1$?